Auditable Policy-Governed LLM-Agent Control for O-RAN Edge Security

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***Abstract*—Open Radio Access Network (O-RAN) security management can use large language models to assess evidence, but their recommendations require external authority and a reproducible audit trail. Our framework treats model outputs as proposals, separates deterministic policy from orchestration, and records evidence and operator approval. TLC checks six invariants over 320 reachable states of a finite policy-gate abstraction. On six synthetic scenarios crossed with three retrieval modes and ten repeats, the proposed Agentic-RAG workflow has 100.0% decision agreement and 96.7% action-class agreement with a frozen experimental oracle; the single-assessor LangGraph configuration has 83.3% and 66.7%, respectively. The workflows differ in prompts, provider calls, request budgets and governance, preventing component-level attribution. A retained-output ablation shows that suppressing execution declarations can leave semantic agreement unchanged. Median decision durations of 80.453–109.793 s restrict intended use to deliberative management, subject to task-specific deadline validation. Public-testnet anchoring measures first inclusion, while one live approval demonstration writes only to a simulated ledger. The evidence supports auditable governance under these experimental conditions; it establishes neither independently validated O-RAN correctness nor production deployment safety.**

***Index Terms*—O-RAN, zero trust, agentic AI, policy governance, auditability, human in the loop, LangGraph, n8n, reproducible evaluation.**

I. Introduction

OPEN Radio Access Network (O-RAN) introduces programmability by disaggregating control and management functions across the Service Management and Orchestration (SMO), Non-Real-Time RAN Intelligent Controller (Non-RT RIC), Near-Real-Time RIC, rApps, xApps, O-CU, O-DU, and O-RU. This flexibility enables closed-loop optimisation and intelligent security operations, but it also creates operational risk: autonomous agents may recommend unsafe actions, bypass mobile-network-operator (MNO) policy, replay historical audit records, or submit unverifiable evidence. The central research question of this paper is: *how can agentic O-RAN security workflows be made policy-gated, evidence-preserving, replay-resistant, and scientifically reproducible without overclaiming production deployment?* Existing O-RAN monitoring and xApp evaluation approaches often focus on telemetry, control logic, or performance outcomes. In contrast, this work treats autonomous O-RAN security control as a governance and accountability problem. The system must not only decide; it must justify, bind, test, and reproduce every decision.

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A. From ZKTrustLLM (ICC Workshops 2026) to This Work

Our earlier conference paper, ZKTrustLLM [26], introduced a three-plane accountability architecture for secure 5G/O-RAN edge multicast: a latency-preserving DTLS/RTP media plane, an IPFS-based cold-path evidence plane referenced through content identifiers (CIDs), and a permissioned-blockchain trust plane that records Groth16-compatible [7], commitment-bound zero-knowledge submissions and couples aggregated trust scores to Logical Key Hierarchy (LKH) rekeying and subgroup-isolation policies. A 10-tenant local prototype (50 submissions, co-located IPFS) showed predictable confirmation latency (p50 = 6 ms, p95 = 7 ms) with stable mean gas of approximately 137,783, demonstrating that commitment-bound accountability can be added without perturbing the real-time media path. That conference design left six gaps addressed by the present journal work:

1) *No agentic control governance:* five specialised roles and the four-class AUTOMATIC–HUMAN–PRIVILEGED–NEVER ladder provide fail-closed rejection.
2) *No formal verification:* an executable TLA$^+$/TLC policy-gate model now exhaustively checks six safety invariants over all 320 reachable states of the configured finite abstraction.
3) *Local-chain evidence only:* a paired, three-session public-testnet study executes identical bytecode and 90 matched pairs on Ethereum Sepolia (Gasper PoS) and IoTeX testnet (Roll-DPoS), complemented by three clean sessions on real three-member etcd/Raft and four-validator Besu/QBFT clusters and a validated SHA-2 hash-primitive comparison.
4) *No semantic LLM evaluation:* a 180-cell frozen synthetic-policy oracle benchmark compares No RAG, RAG, and Agentic RAG, adds an external LangGraph baseline and a local open-weights run, and reports clustered uncertainty.
5) *No runtime negative-security evidence:* replay, zero-anchor, and unauthorised-submitter rejection is now evidenced in local hooks and on both public testnets, with explicit claim-boundary validation.
6) *No human-governed live execution:* a live n8n supervisor run demonstrates a HUMAN gate terminating in a side-effect-free simulated ledger update, cross-bound to machine-verifiable report and action hashes, plus a reviewer-oriented reproducibility hardening layer.

The proposed system addresses the research question by combining agentic reasoning, zero-trust policy gates, content-

addressed evidence, smart-contract audit anchoring, and explicit claim-boundary validation. The novelty is not the isolated use of blockchain, IPFS-style content identifiers, n8n, or LLM-assisted reasoning. Instead, the contribution is their controlled integration into an auditable framework for accountable agentic O-RAN security. The design is motivated by zero-trust principles [1], O-RAN WG11 security requirements and threat models [3], [4], [5], 3GPP 5G security architecture [6], O-RAN policy-control workflows [2], xApp development practice [13], workflow orchestration [14], and recent agentic AI-RAN research [9], [10], [11], [12].

B. Contributions

The paper makes the following contributions:

- *Governed architecture:* a two-plane design coupling O-RAN governance/edge functions to an agentic decision plane, with a deterministic four-class action ladder, hash/CID evidence binding, and smart-contract anchoring that rejects replayed, zero, and unauthorised submissions.
- *Formal and reproducible pipeline:* an executable TLA$^+$/TLC policy-gate model (six invariants over 320 reachable states) and an n8n-orchestrated pipeline producing a 240-record local evidence dataset with runtime negative-security evidence.
- *Consensus and integrity evidence:* a paired, three-session study executing identical audit-anchor bytecode over 90 matched pairs on Ethereum Sepolia (PoS) and IoTeX testnet (Roll-DPoS), extended by a permissioned etcd/Raft and Besu/QBFT fault study and a validated Keccak-256/SHA-256/SHA-512 hash-primitive microbenchmark.
- *Frozen semantic evaluation and live governance:* a 180-cell oracle benchmark with scenario-clustered intervals, an external LangGraph baseline, a local Qwen3-4B run, a 30-case prompt-injection suite, and a live HUMAN-gated execution backed by an append-only 233-attempt index.

C. Organization

Sections II–V position the work, define the system and threat model, present the two-plane architecture, and analyse the policy gate formally. Sections VI and VII describe the reproducible evaluation and its results across the frozen oracle, live governance workflow, ledger studies, security tests, and claim boundary. Section VIII concludes and identifies the next validation steps.

II. Related Work and Positioning

Zero trust reduces implicit trust by enforcing accurate per-request access decisions in environments that may contain compromised segments [1]. It applies directly to O-RAN, where xApps, rApps, RIC components, telemetry collectors, audit services, and orchestration modules span different trust domains: no autonomous action should be trusted merely because it originates from an agent. Zero-trust RIC frameworks and xApp prototypes [27], [28], machine-learning-driven zero-trust designs for 5G/6G and O-RAN [34], and surveys of zero-trust and blockchain-assisted architectures for 6G and IoT [29], [30] chart this design space but do not govern LLM-agent actions or bind decisions to reproducible evidence. O-RAN enables policy-driven control through SMO, Non-RT RIC, Near-RT RIC, A1, E2, O1, and O2 interfaces [2], [3], [4]. WG11 threat modelling and 3GPP TS 33.501 motivate explicit trust boundaries, authorisation, integrity, and audit controls [5], [6]. A1 is the intended operational locus here: model reasoning is separated from the deterministic policy and audit functions that govern a candidate policy change. Agentic AI is increasingly proposed for RAN reasoning, security compliance, and network automation [9], [10], [11], [12], supported by a broader literature on LLMs for telecommunications and network/service management [31], [32], [33] and by general-purpose multi-agent frameworks such as CrewAI, AutoGen, and CAMEL [17], [18], [19]. These systems can interpret intent and coordinate tools, but they leave an accountability question: what evidence allows an operator or reviewer to reconstruct, reject, and reproduce a decision? Our contribution is an external action ladder, replay-resistant evidence binding, negative-security tests, and a frozen evaluation protocol. The most directly related security-compliance study is the standards-aware agentic AI framework of Chatzimiltis *et al.* [9], which integrates LLM-based agents with retrieval-augmented generation to inspect configuration artifacts against O-RAN Alliance and 3GPP requirements, produce explainable compliance judgements, and propose remediation. That study establishes the motivation and architectural role of autonomous RAN security-compliance agents. The proposed system addresses a complementary accountability gap: candidate judgements are constrained by an explicit four-class action ladder and hash/CID evidence binding, with replay, zero-anchor, and unauthorised-submitter rejection. n8n is an orchestration layer rather than a security primitive; policy enforcement remains outside the model and workflow engine [14]. Consensus is a property of the execution network, not of the Solidity contract. Ethereum documents Sepolia as an application-development testnet that follows the Ethereum protocol while using a permissioned validator set controlled by client and testing teams [20]. IoTeX documents Randomized Delegated Proof of Stake (Roll-DPoS), which combines delegated candidate election, randomized committee selection, and PBFT-style finalisation [21]; its EVM-compatible testnet is identified by chain ID 4690 [22]. These network properties motivate a controlled sensitivity experiment, but they do not by themselves justify a causal claim about consensus performance. The present study therefore holds the contract bytecode and paired transaction payloads constant, reports client-observed first inclusion, and treats RPC behaviour, propagation, block cadence, load, and consensus as inseparable contributors to the observed public-testnet latency.

A. Implementation Testing and Runtime Assurance

ORANalyst [35] combines dependency analysis and dynamic tracing to test O-RAN implementations. InvaRAN [36] uses inferred invariants to expose controller logic flaws. Their implementation evidence is different from checking an abstract proposal gate. VeraRAN [37] models asynchronous multi-interface actions and repairs event dependencies; it addresses intermediate-state safety that our finite gate does not model. The

TABLE I
RELATED WORK BY ASSURANCE TARGET. THIS IS A SCOPE COMPARISON, NOT A CLAIM THAT UNREPORTED CAPABILITIES ARE ABSENT.

Study	Primary assurance or management target
ORANalyst [35]	Dependency-aware implementation testing
InvaRAN [36]	Invariant-guided controller logic testing
VeraRAN [37]	Asynchronous plan safety and event ordering
Safety Copilot [38]	Runtime supervisory verification
Multi-layer defence [39]	Near-RT message, telemetry and xApp safeguards
Salmi *et al.* [40]	Conflict-aware control and intent orchestration
This work	Oracle-relative recommendation evaluation and auditable proposal governance; simulated actuation

TABLE II
COMPONENT STATUS. AN IMPLEMENTED WORKFLOW MAY TERMINATE IN A SIMULATED ACTION.

Component	Evidence status
Assessment, reconciliation, audit	Implemented experimental workflow
HUMAN review callback	One observed approved simulation
Action target	Simulated ledger; no network side effects
Policy-gate invariants	Finite abstraction model checked
Oracle evaluation	Six synthetic scenarios; no expert validation
Public anchoring	Testnet first-inclusion experiment
SMO/rApp/A1/E2 deployment	Intended; not implemented in this evaluation

Safety Copilot [38] studies independent runtime supervision of AI-driven control, and the multi-layer defence framework of Alimohammadi *et al.* [39] integrates E2 inspection, telemetry-poisoning detection and xApp attestation. These are relevant enforcement precedents; the present work makes no priority claim for external runtime supervision. Salmi *et al.* [40] investigate AI-native O-RAN conflict management and intent orchestration. Our narrower contribution is the recorded linkage between evidence, untrusted assessment, policy routing and approval in a reproducible management workflow. Table I separates these evidence targets.

III. System and Threat Model

A. Target Scenario

The target deployment is an O-RAN edge environment with MNO governance, SMO/Non-RT RIC policy management, Near-RT RIC xApps, O-CU/O-DU/O-RU components, and media or telemetry flows represented by RTP/DTLS-RTP impairment profiles. The framework is designed for security-operations accountability and scientific validation, not as a production replacement for a RIC, SMO, or real-time RAN control loop.

B. Agentic Components

Five functions are separated. The telemetry function observes O-RAN/media KPIs and impairment profiles; the reasoning function maps fixed features to a trust state and candidate action; the policy function enforces the action ladder; an optional AUTH component records component-level admissibility experiments; and the audit/anchor function binds CID/hash records to commitments. A reasoning model can propose an action, but it cannot bypass the external policy gate or authorise an audit record.

C. Threat Model

The adversary may attempt to: T1) trigger an unapproved autonomous action; T2) tamper with evidence; T3) replay a previous anchor; T4) submit a zero or empty anchor; T5) escalate to a forbidden action; or T6) act as an unauthorised submitter. The framework assumes that cryptographic primitives are implemented correctly and that private keys are not deliberately leaked by a trusted operator. The smart contract is not assumed to know physical network truth. It only enforces trust-plane constraints over commitments, duplicate submissions, zero commitments, role/capability checks, and negative-security invariants. The threat model follows O-RAN WG11 Security Requirements and Controls, Security Protocols, and Threat Modeling specifications [3], [4], [5], together with 3GPP TS 33.501 Release 18 [6]. Retrieved passages and scenario fields are untrusted data: they may contain prompt-injection text attempting to forge approval, invert policy, or induce automatic execution. Fixed feature extraction, frozen evidence hashes, schema validation, deterministic reconciliation, and the external ladder reduce this risk; they do not prove resistance to arbitrary adversarial inputs.

IV. Proposed Architecture

Fig. 1 presents the proposed architecture. It couples two planes: the O-RAN governance/edge environment and a deliberative agentic control path. MNO policy enters through the SMO/Non-RT RIC and A1 path; model output remains a proposal until deterministic governance assigns an action class. Evidence services preserve the input, decision, and outcome lineage.

A. Implemented and Intended Integration

The evaluated input is a JSON configuration with named O-RAN components and retrieved policy evidence; it is not a standard E2 indication or a live telemetry stream. The output is a candidate decision and action class with hash-bound evidence. The reported prototype implements reconciliation, policy routing and audit recording, but its action target is simulated. Fig. 1's A1/E2 path identifies intended integration. No deployed rApp, standard A1 policy exchange, or live E2 actuation is demonstrated. An operational adapter would need schema-validated A1 policy instances, authenticated SMO authority, fresh target state and independently observed application outcomes. Those are future integration requirements.

B. Policy Ladder and Threat-Response Mapping

The policy-gated control path is summarised in Table III, which replaces the earlier decision-loop and threat-response diagrams with a compact comparative structure. The central principle is that reasoning only proposes candidate actions; execution is mediated by the action ladder and audit anchoring. The table also maps threats to the enforcement component and accountable outcome.

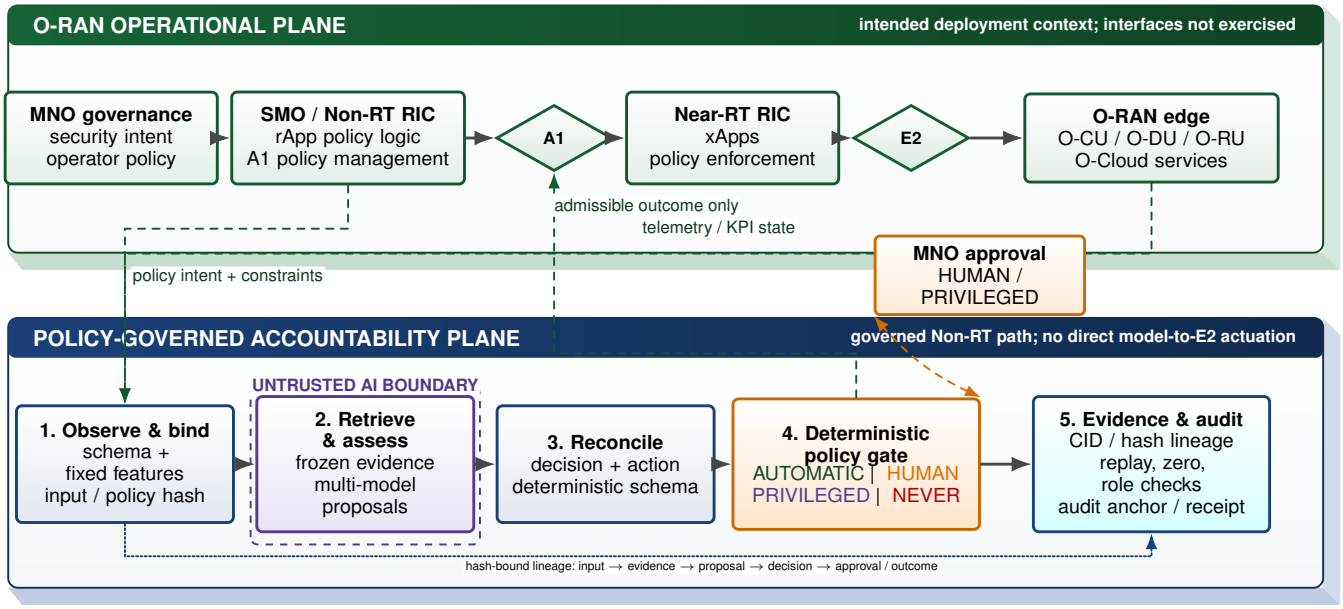


Fig. 1. Two-plane architecture for policy-governed O-RAN security. The operational plane depicts the intended standard SMO/Non-RT RIC–A1–Near-RT RIC–E2 path. The accountability plane binds policy and telemetry, isolates model assessment as untrusted, reconciles proposals, applies the four-class gate, and records the audit lineage. The SMO/A1 return is architectural and was not exercised; HUMAN and PRIVILEGED actions require MNO approval. Models never actuate E2 directly.

TABLE III
POLICY LADDER AND THREAT-RESPONSE MAPPING.

Case	Input / condition	Enforcement component	Evidence path	Accountable outcome
OBSERVE–REASON	Telemetry, KPI, or profile state produces a candidate action	Telemetry and reasoning agents	Run identifier, profile metadata, local artifact state	Candidate trust state and action proposal; no direct actuation
AUTOMATIC	Low-risk action that is policy-admissible	Policy and schema gates	Action class, admissibility check, audit record	Executed only if admissible and audit-bound
HUMAN	Action requiring operator judgement	Policy ladder	Approval queue and recorded reason	Queued for human approval; not automatically executed
PRIVILEGED	Action requiring stronger authority	Privileged approval boundary	Privileged approval record and trace	Blocked from automatic execution; requires out-of-band approval
NEVER	Forbidden or unsafe action	Hard policy rejection	Rejection reason and audit trace	Blocked by construction
Evidence tampering	Modified evidence object or mismatched pointer	Hash/CID and audit path	Content-addressed evidence and hash-chain continuity	Tampered evidence is detectable and not silently trusted
Replay attempt	Duplicate prior commitment	Smart-contract duplicate check	Existing commitment and rejection event	Replayed commitment is rejected
Zero-anchor attempt	Empty or zero commitment	Smart-contract invariant	Non-zero commitment check	Zero commitment is rejected
Unauthorised submitter	Submitter lacks authority	Role/capability gate	Submitter identity and rejection event	Unauthorised submission is rejected

C. Smart-Contract Boundary

The smart contract does not control radio traffic directly. Its role is limited to audit accountability: storing compact commitments, rejecting duplicate commitments, rejecting zero commitments, and enforcing authorised submitters. It cannot guarantee IPFS availability, prove physical network truth, or replace MNO operational policy. This boundary avoids overclaiming: the ledger records and verifies governance evidence, while the O-RAN control plane remains responsible for real network execution.

The archived HUMAN demonstration's ANCHORED status denotes a local hash-chain record: its external-chain evidence field is MISSING. Its audit record must not be conflated with the separate 90-pair public-testnet experiment. Inclusion of a commitment cannot establish telemetry truth, availability of off-chain evidence, key security or economic finality. Those assumptions remain outside the measured latency study.

V. FORMAL SAFETY ANALYSIS

The executable TLA+ gate has state fields `phase`, `actionClass`, `humanApproved`, `privilegedApproved`, `proofValid`, `anchorCommitment` and `executed`. Its six invariants state: NEVER implies no execution; executed HUMAN requires human approval; executed PRIVILEGED requires both approval Booleans; execution requires abstract admissibility and a non-zero commitment; and execution implies the ACT phase. The corresponding identifiers are `NeverNotExecuted`, `HumanRequiresApproval`, `PrivilegedRequiresBoth`, `ProofRequired`, `NonZeroAnchorRequired` and `NoExecutionBeforeAct`. Exact predicates are given in the supplement. `proofValid` denotes abstract admissibility, not a successfully verified cryptographic proof. Approval Booleans are assumptions about state, not a verified human-authentication protocol.

We reproduced TLC 2.19's original 64 initial states, 320 reachable states, depth five and zero invariant violations [8]. Its initial predicate excludes zero commitments, making the non-zero property unexercised on zero inputs. A separate revision probe admits zero, yielding 96 initial and 480 reachable states with no invariant violation. Three mutations removing admissibility, the non-zero execution guard, or HUMAN approval for PRIVILEGED each produce the expected counterexample. The original model is preserved. It contains no history of accepted commitments: duplicate-anchor rejection is runtime/contract evidence, not a TLC result. Neither run proves liveness, code refinement, model semantic correctness, asynchronous RAN safety or production deployment safety. ACT/BLOCK are terminal; `-deadlock` suppresses terminal-state warnings and does not establish termination. Logs and model hashes accompany the revision.

VI. Reproducible Evaluation Methodology

A. Staged Evaluation

Six stages progressively add evidence: (1) the response matrix, KPI plan, executable TLA+ model, and n8n skeleton; (2) the clean 240-record workflow matrix; (3) anchor, reasoning, and negative-security runtime hooks; (4) Sepolia validation; (5) the paired PoS–Roll-DPoS and local hash-primitive studies; and (6) three real etcd/Raft and Besu/QBFT permissioned-cluster sessions. The paper harness additionally performs packet-capture smoke testing, AUTH_V2.2 timing, AI-evidence diagnostics, Stage F/F row hashing, and evidence-manifest generation.

B. Paired PoS–Roll-DPoS Consensus-Sensitivity Study

The Stage 5 experiment asks a deliberately bounded question: when identical `Stage3AnomalyLedger` runtime bytecode receives matched audit-anchor transactions at approximately the same time, how do client-observed first-inclusion latency, EVM execution gas, and observed block cadence differ between Ethereum Sepolia and IoTeX testnet? Sepolia uses chain ID 11155111 and the Ethereum PoS protocol, but its permissioned test validator set prevents extrapolation to Ethereum mainnet [20]. IoTeX testnet uses chain ID 4690, exposes an Ethereum-compatible JSON-RPC endpoint, and runs Roll-DPoS [21], [22]. The Solidity 0.8.20 contract is compiled once (optimizer 200, `viaIR=true`). Each session verifies both chain IDs, deploys the same creation bytecode, checks non-empty deployed code, and requires identical runtime-code Keccak-256 hashes across networks and sessions. After three excluded warm-up pairs, each session executes 30 measured pairs; pair (s, i) carries the same commitment and evidence digest on both networks, submitted concurrently from the same client host, so three sessions provide $N = 3 \times 30 = 90$ pairs. The client host, per-network RPC provider, compiler settings, bytecode, gas limit, and one-confirmation receipt rule remain fixed. For network $n \in \{S, I\}$, where S denotes Sepolia and I denotes IoTeX, the primary observation is

$$L_{s,i}^{(n)} = t_{s,i,\text{receipt}}^{(n)} - t_{s,i,\text{submit}}^{(n)}. \tag{1}$$

This interval begins immediately before the JSON-RPC transaction submission and ends when a successful receipt is available after one inclusion. It includes client/RPC overhead, propagation, producer waiting time, and prevailing public-testnet load; it is not economic-finality latency. The paired estimands use the same orientation:

$$\Delta_{s,i} = L_{s,i}^{(S)} - L_{s,i}^{(I)}, \qquad R_{s,i} = \frac{L_{s,i}^{(S)}}{L_{s,i}^{(I)}}. \tag{2}$$

Both estimands are summarised by their medians over pairs; the reported ratio is therefore a median of per-pair ratios and is not expected to equal the quotient of the two marginal medians. The harness also records deployment and anchor gas, effective gas price, transaction and block identifiers, receipt/event integrity, and at least 64 consecutive post-run block headers per session. Duplicate, zero-commitment, and unauthorised-submitter checks run as read-only `eth_call` probes against each deployed state, testing rejection semantics without mining deliberately reverting transactions; native-token gas prices are retained for provenance only. The deterministic analysis reports medians, quartiles, P95, means, and sample standard deviations using R-7 linear-interpolated quantiles. Median intervals use 10,000 replicates of a two-level hierarchical bootstrap with fixed seed 20260721: sessions are resampled first and matched pairs are resampled within each selected session. This preserves the session-clustered design. The publication gate fails closed for fewer than three sessions, fewer than 30 measured pairs per session, wrong chain IDs, bytecode mismatch, unsuccessful receipts, missing anchor events, or any failed rejection probe. Each raw `benchmark.json` has a companion `SHA256SUMS.txt`; a derivation manifest binds the three input hashes to the generated summary, table, figure, and results paragraph. Because only three session clusters are available, the bootstrap intervals are descriptive summaries of the retained sessions; reliable nominal 95% coverage is not claimed.

C. Permissioned Raft–QBFT Fault Experiment

Pinned etcd 3.5.17 and Besu 24.5.4 clients run a real three-member Raft cluster and four-validator QBFT network, respectively [24], [25], [23]. Three clean sessions retain 30 observations per latency condition ($n = 90$). Raft is tested healthy, with one follower stopped, after leader failover, without a majority, and after restoration. QBFT is tested healthy, with 3/4 and 2/4 validators, and after operator-assisted restoration; the 2/4 case records bounded block-height probes, not transaction latency. Publication requires correct identities and counts, expected progress/non-progress, contract/event integrity, clean Git state, and valid SHA-256 manifests. Because Raft measures etcd key/value replication while QBFT executes `Stage3AnomalyLedger`, latency is compared only within each family. Injected faults are process crash/non-participation and quorum loss, not equivocation, forged votes, state corruption, or arbitrary Byzantine messages.

D. Frozen Synthetic-Policy Oracle Benchmark

The retrieval comparison is evaluated through a separate, frozen oracle benchmark rather than being merged with the 240-record impairment-profile matrix. The benchmark crosses six synthetic O-RAN policy scenarios with three retrieval modes—No RAG, RAG, and Agentic RAG—and ten recorded repetitions, giving $6 \times 3 \times 10 = 180$ canonical successful cells. Each cell is evaluated against the same executable synthetic operator-policy oracle. Freezing the oracle, prompts, scenario payloads, and metric definitions before aggregation prevents post-hoc relabelling; the oracle is an experimental reference function, not expert-verified ground truth or a claim of comprehensive O-RAN/3GPP compliance. The execution design separates semantic population from operational attempts: the semantic denominator contains one successful package per planned cell, while an append-only attempt index retains every invocation, including failed workflow-boundary attempts, and SUCCESS-only resumption retries incomplete cells without deleting prior failures. The 233 operational attempts (180 successes, 53 retained HTTP-500 failures) therefore measure workflow reliability and resumability, whereas the 180 successful cells measure oracle-relative behaviour; $180/233$ is not model accuracy or a final-cell completion rate. Whole-scenario percentile intervals use 20,000 fixed-seed bootstrap replicates over the six scenario clusters. Repeats assess within-cell stability, not independent scenario-level generalisation. Independent human validation was not conducted because eligible annotators were unavailable; the frozen oracle will not be revised in response to any future labels. Fig. 2 presents the five-stage path used for the oracle benchmark and live demonstration: observe, propose, govern, bind, and audit. The supplement retains the complete exported canvas, node identifiers, and archived execution trace.

For each retrieval mode m, let $N_m = 60$, let d_i be the final decision, a_i the enforced action class, and $(d_i^\star, a_i^\star)$ the frozen oracle labels. Let $z_i = 1$ denote a covered, non-abstaining output. The principal oracle-relative metrics are

$$\mathrm{AD}(m) = \tfrac{1}{N_m}\textstyle\sum_i \mathbf{1}[d_i = d_i^\star], \quad \mathrm{AA}(m) = \tfrac{1}{N_m}\textstyle\sum_i \mathbf{1}[a_i = a_i^\star], \tag{3}$$

$$\mathrm{C}(m) = \tfrac{1}{N_m}\textstyle\sum_i z_i, \qquad \mathrm{AS}(m) = \dfrac{\sum_i z_i\,\mathbf{1}[d_i = d_i^\star]}{\sum_i z_i}. \tag{4}$$

Here, AD is decision agreement, AA is enforced-action agreement, C is coverage, and AS is selective decision agreement. Abstention is $1 - \mathrm{C}$. Safe-given-abstention is calculated over abstaining cells and reports whether the enforced route remained fail-closed. If $\sum_i z_i = 0$, selective agreement is undefined and reported as N/A. Provider and role-chain validation are deterministic artifact checks and are reported separately from oracle-relative semantic agreement.

a) Scenario construction and limits: The frozen configurations are author-designed policy cases, not sampled operator incidents. Table IV maps their exact fields to O-RAN management concerns. P-01–P-04 are clauses of an explicitly experimental operator policy, not normative O-RAN clause identifiers. The oracle assigns HUMAN even to compliant S1: its label encodes a review policy, not permission for autonomous execution. None of the six reference classes is AUTOMATIC or PRIVILEGED. The benchmark therefore does not assess useful automatic execution or the full four-class distribution. Missing evidence is explicit in S6, and S2/S4 contain a request conflicting with policy. Retrieved distractors and the two reconciliation mismatches are recorded, but this is not a systematic noisy-telemetry or adaptive-conflict study. The separate delay/jitter/loss experiment configures workflow stress; it does not measure radio-channel realism. Independent expert labels, production-derived scenarios and held-out noise/conflict validation remain absent. We narrow the conclusions accordingly and preserve the frozen oracle.

TABLE IV
FROZEN SCENARIOS AND ORACLE LABELS. C/NC/U DENOTE COMPLIANT, NON_COMPLIANT AND UNCERTAIN; H/N DENOTE HUMAN AND NEVER.

ID	Configuration and management concern	Decision	Class
S1	O-CU/O1; NEA2, NIA2, integrity enabled, TLS 1.3	C	H
S2	O-CU/F1-U; integrity disabled despite required MNO profile (P-01)	NC	H
S3	SMO–O-CU/O1; TLS 1.0 compatibility request (P-02)	NC	H
S4	Near-RT RIC/E2; unregistered xApp, RBAC false (P-03)	NC	N
S5	Audit service; duplicate prior commitment (P-04)	NC	N
S6	O-DU; unverified summary, unknown interface, absent security profile	U	H

E. Artifact Reproducibility

The historical public artifact is version 1.0.0 [23]. The revision-specific reproducibility archive adds the marked/clean sources, unchanged retained-data packages, reanalysis code, formal logs and editable figures. Its build information identifies the repository commit and file digests. These new checks must not be attributed to the older DOI release. The supplement describes the evidence objects and reproduction commands. The revision includes all 180 recovered proposed-workflow report packages, with 180 manifests binding 3,240 evidence files, plus retained baseline records. Every recovered report's canonical hash matches its entry in the previously released frozen oracle export. Reanalysis reproduces the agreement counts, scenario-clustered intervals and indexed latency statistics without new inference. This establishes internal integrity and result consistency; it does not independently authenticate provider services or validate the synthetic oracle.

F. Claim Boundary

Claims are restricted to the released implementation and retained evidence. TLC establishes six invariants only for the finite model and explored state space. Oracle metrics measure agreement with a frozen synthetic reference, not expert-verified or population-level accuracy. Public-testnet results measure client-observed first inclusion and include RPC, propagation, cadence, load, and chain-policy effects; they do not measure economic finality or isolate consensus. The permissioned faults cover crash/non-participation, not arbitrary Byzantine behaviour. The live n8n path uses simulated actuation and a simulated ledger and does not establish production O-RAN integration,

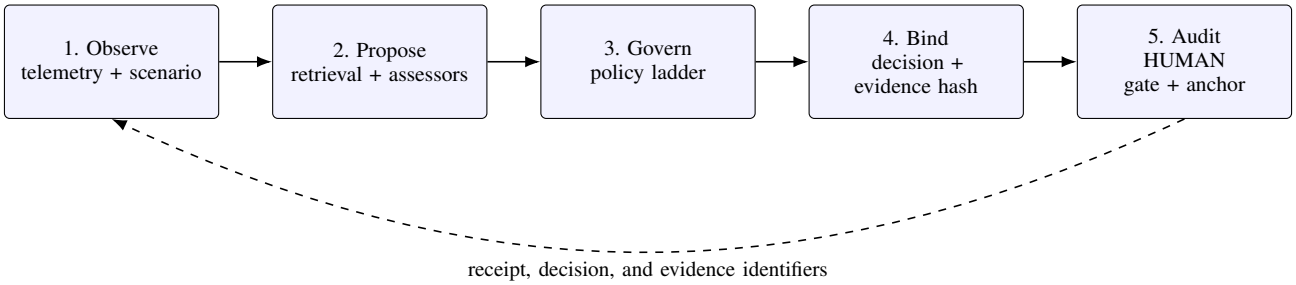


Fig. 2. Five-stage execution schematic. Model output remains untrusted until the external policy ladder admits it. The complete exported workflow and its node-level configuration are provided in the supplement.

TABLE V
EVALUATION MATRIX AND EVIDENCE OUTPUTS.

Dimension	Values
Variants	Full system, no AUTH, oracle-only, RBAC-only, no IPFS, no policy gate
Profiles	Clean, delay, delay+jitter, delay+jitter+loss
Repeats	10 per variant/profile pair
Core records	$6 \times 4 \times 10 = 240$ local clean records
Oracle benchmark	6 scenarios $\times$ 3 retrieval modes $\times$ 10 repeats = 180 successful cells
External baseline	LangGraph: the same 180 frozen cells; 180/180 HTTP 200
Open weights	Qwen3-4B: 60 Agentic-RAG cells; local Ollama endpoint
Injection suite	10 attack families $\times$ 3 base contexts = 30 cases
Operational attempts	233 append-only attempts: 180 successful and 53 retained HTTP-500 failures
Consensus sensitivity	3 sessions $\times$ 30 measured pairs = 90; 9 warm-up pairs excluded
Permissioned faults	3 sessions; 90 retained observations per latency condition and 90 quorum-loss probes per implementation
Block-cadence sample	189 consecutive intervals per network across the three sessions
Hash microbenchmark	3 primitives $\times$ 2 payload sizes $\times$ 30 retained `eth_call` observations (3 warm-ups excluded per cell)
Formal verification	TLC 2.19; 64 initial states, 320 distinct reachable states, depth 5, six invariants
Outputs	Run ID, Git hash, raw logs, evidence hashes, KPI JSON/CSV, claim-validation report, evidence manifest

availability, or safety. Cross-provider observations apply to the recorded aliases, prompts, and access dates and need not persist after provider updates. The frozen 240-row matrix has no per-row proof outcome or proof/public-input binding and therefore supports 0/240 row-bound successful verifications. AUTH_V2.2 is reported only as a component experiment, including its observed `verifyTx=false` outcome; no system-level zero-knowledge claim is made. Prompt-injection and open-weights results are likewise confined to their specified synthetic suites. n8n coordinates experiments but remains outside the trusted computing base. Table V summarises the denominators.

G. AI-Assisted n8n Workflow Extension

Additional Stage E/F and Stage F/F orchestration observations are moved to the supplement. Their small, selectively valid samples are not used for the principal effectiveness or generalisability claims. The oracle benchmark uses GPT-5.6 as an assessor; the distinct GPT-5.5 alias appears only as the planner in the supplemental Stage F/F workflow.

VII. Results and Discussion

a) Operational timescale and intended use: Near-RT RIC loops operate around 10 ms–1 s; Non-RT rApp loops exceed 1 s [2]. The recorded 80.453–109.793 s medians and 164.747 s Agentic-RAG P95 exclude Near-RT control and urgent inline containment. Non-RT positioning alone does not validate a deadline. Candidate tasks are policy-change preparation, compliance review and scheduled remediation planning, subject to an operator-approved budget above the observed tail and evidence-freshness checks. No task-specific service-level objective is validated here. Human waiting follows recommendation generation and must be budgeted separately. When a deadline or evidence lifetime expires, a deployment should withhold actuation and require fresh assessment; that expiry behaviour is a requirement, not an observed production property.

A. Live n8n Execution and Human-Governed Outcome

To distinguish architectural intent from observed execution, the final-paper experimental workflow was exercised as a live supervisor demonstration using scenario S2 (DRB integrity disabled) in AGENTIC_RAG mode. The production webhook returned HTTP 200, and n8n recorded successful master execution #1412 in 109.436 s for run `d7c0e31b-fa9e-437b-87d2-acdeef4a6ba1`, with a NON_COMPLIANT decision and the HUMAN action class enforced. Four of four provider stages and four of four agent-role stages passed the artifact checks, after which the report reached the ANCHORED state. Execution #1412 traverses the five-stage workflow of Fig. 2: input binding, retrieval, independent provider assessment, consensus, the planner–critic–recovery–arbiter chain, policy and formal-evidence gates, audit anchoring, and governed-action preparation. (The n8n UI shows host-local BST while evidence files use UTC; the one-hour offset is expected and does not denote a second run.) Fig. 3 isolates the governance transition. A separate callback first validates the reviewer payload, then submits the authorised decision to the gateway. Execution #1417 completed successfully in 39 ms and followed the *true* branch of *Review Approved?*; the rejection branch remained unexecuted. The gateway subsequently recorded APPROVE and SIMULATED_EXECUTED for the governed action. Crucially, approval did not authorise production actuation: the execution target was `SIMULATED_O_RAN_ACTION_LEDGER`, `production_oran_execution` was `false`, and

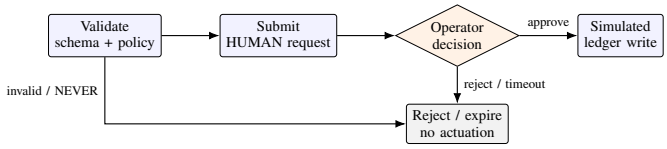


Fig. 3. HUMAN-gate state transition. The approved simulated-ledger branch was observed. Invalid, reject and timeout branches show intended fail-closed requirements; complete branch coverage is not claimed.

`network_side_effects` was `false`. The split between the main workflow and review callback therefore operationalises the HUMAN class as a distinct, auditable control transition rather than an implicit continuation of model reasoning.

The visual trace is cross-bound to machine-verifiable artifacts rather than treated as primary proof. The full report and governed-action SHA-256 digests are retained in the artifact; their prefixes/suffixes are `8a36aed6...f4b65e9` and `4a345bb1...f7124d18`. Fourteen consistency conditions over the report and review response passed, and the evidence directory includes a recursively verified SHA-256 manifest. Consequently, the machine-readable artifacts establish that the documented workflow executed and traversed the HUMAN gate. Its interpretation is governed by the Claim Boundary in Section VI-F.

a) Operator authority and review boundary: The archived callback records a reviewer identifier, claimed role, Boolean decision and rationale, with run and report hashes linking the pending action to its outcome. The HUMAN role is an MNO approver; PRIVILEGED denotes an additional authority requirement, with both approval Booleans required by the formal abstraction. A recorded identifier/role is not evidence of cryptographic reviewer authentication. The retained positive callback and simulated write establish one transition only: reject, expiry, replay, concurrent decisions and approval revocation are not empirically established.

A deployment approval view should expose the exact action and target, policy version, supporting evidence and uncertainty. An authenticated decision should bind the action/evidence version and include identity, authority, timestamp, reason and expiry. Conflicting decisions, stale evidence or changed payloads should leave the action blocked until a fresh authorised decision. These remain deployment requirements. A new inactive callback export corrects Boolean string coercion with strict validation; local payload tests do not demonstrate gateway authentication or deployed approval enforcement.

B. Frozen Oracle Benchmark Performance

Table VI reports 180 canonical successful cells, 60 per retrieval mode. No RAG yields 16.7% decision agreement, 66.7% action agreement and zero coverage: every cell abstains. Its 100% safe-given-abstention rate describes fail-closed gating under the reported configuration; it does not establish semantic correctness.

RAG raises decision and enforced-action agreement to 83.3%, with 66.7% coverage and 33.3% abstention. Agentic RAG reaches 100.0% decision agreement, 96.7% enforced-action agreement, and 83.3% coverage. Both retrieval modes achieve 100.0% selective agreement on covered cells and 100.0% safety conditional on abstention. Therefore, the gain is not obtained by forcing a decision in every case: the workflow retains a measurable abstention channel and keeps abstaining cases fail-closed. Decisions were repeat-stable in all 18 scenario–mode groups; enforced actions were stable in 16/18. Agentic-RAG S2 produced HUMAN in nine repeats and NEVER in one, while S4 produced NEVER in nine and HUMAN in one. Ten repeats do not create 60 independent scenarios. With only six clusters, the 20,000-replicate intervals are descriptive and reliable nominal coverage is not claimed. Fig. 4(a) visualises the principal semantic metrics. The difference between Agentic-RAG decision agreement (100.0%) and action agreement (96.7%) is important: the final semantic decision matched the oracle for every cell, but deterministic evidence reconciliation changed the enforced class in two of 60 cells. Fig. 4(d) shows the associated latency cost. Relative to No RAG, RAG increases median latency by 19.699 s (24.5%) and P95 latency by 42.016 s (38.6%). Agentic RAG increases the median by a further 9.641 s and P95 by 13.978 s. End-to-end, Agentic RAG is 36.5% slower at the median and 51.5% slower at P95 than No RAG. The benchmark therefore supports an agreement–coverage–latency trade-off rather than an unqualified claim that the most agentic mode is superior for every operational objective.

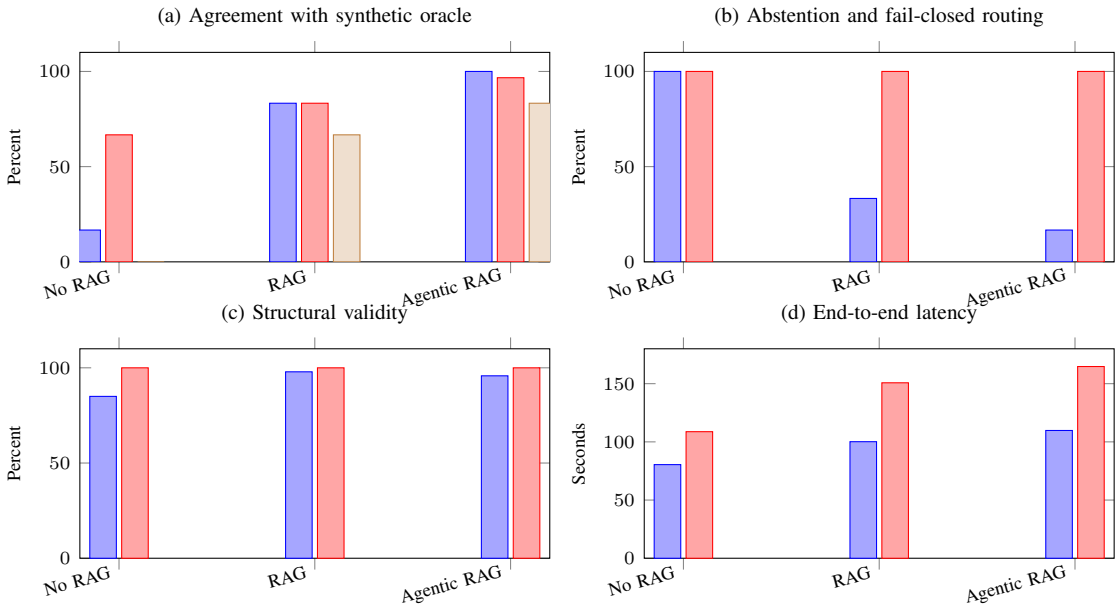
Table VII diagnoses the two Agentic-RAG action-class mismatches. In S2 repeat 10, unsupported P-03 evidence was retained together with P-01, so deterministic reconciliation escalated HUMAN to NEVER. This error is over-restrictive and preserves safety because automatic execution remains disabled. In S4 repeat 5, exact P-03 evidence was not retained and the final clauses became UNSPECIFIED, so reconciliation downgraded the model proposal from NEVER to HUMAN. Although this direction is under-restrictive relative to the oracle, HUMAN still requires explicit review and therefore remains fail-closed. These two cases explain the $58/60 = 96.7\%$ action agreement while automatic execution remained disabled in this benchmark.

C. External and Open-Weights Baselines

The external LangGraph [15] baseline used version 1.2.10, the same canonical 180-cell JSONL (SHA-256 `4854f1d4...e1196c6c`), the same scenario labels and scoring, and one assessor node followed by deterministic reconciliation. All 180 requests returned HTTP 200. Temperature was 0, top-p was 1, and the completion limit was 256 tokens; the endpoint returned the requested `gpt-5.6-terra` alias. Table VIII replaces a qualitative tick comparison with measured results. For all systems, NEVER recall is the fraction of oracle-NEVER cases whose final action class is NEVER; it is distinct from an execute-true flag. The recovered raw reports and retained repeat counts give proposed Agentic-RAG recall $19/20 = 95.0\%$: S4 has nine NEVER and one HUMAN result, while S5 has ten NEVER results. The HUMAN mismatch remains an action-class error even with automatic execution disabled. This corrects the previous table's 100.0% entry. The LangGraph baseline recorded zero NEVER-boundary bypass flags and zero unsafe execution declarations; its estimated API

TABLE VI
AGREEMENT WITH THE FROZEN SYNTHETIC ORACLE AND PROPOSED-WORKFLOW COVERAGE. EACH MODE HAS SIX SCENARIOS AND TEN REPEATS ($n = 60$). BRACKETED SCENARIO-CLUSTERED BOOTSTRAP INTERVALS ARE DESCRIPTIVE; THEY DO NOT VALIDATE O-RAN CORRECTNESS OR GENERALISATION.

Mode	Decision agreement (%)	Enforced-action agreement (%)	Coverage (%)
No RAG	16.7 [0.0, 50.0]	66.7 [33.3, 100.0]	0.0 [0.0, 0.0]
RAG	83.3 [50.0, 100.0]	83.3 [50.0, 100.0]	66.7 [33.3, 100.0]
Agentic RAG	100.0 [100.0, 100.0]	96.7 [93.3, 100.0]	83.3 [50.0, 100.0]



(a) Decision agreement; action agreement; coverage
(b) Abstention; fail-closed given abstention
(c) Provider validity; role validity
(d) Median; P95

Fig. 4. Consolidated frozen-oracle results (60 cells per mode). Agreement is relative to six synthetic scenarios. Safe routing and structural validation do not establish semantic correctness. Latency is reported separately; no operational deadline is validated, and Near-RT use is excluded.

TABLE VII
AGENTIC-RAG ACTION-CLASS MISMATCH DIAGNOSIS. BOTH MISMATCHES REMAINED FAIL-CLOSED.

Cell	Oracle / enforced	Evidence-boundary cause	Safety outcome
S2, r10	HUMAN / NEVER (over-restrictive)	Unsupported P-03 retained with P-01; reconciliation escalated the action.	OPA enforced NEVER; automatic execution remained disabled.
S4, r5	NEVER / HUMAN (under-restrictive)	Exact P-03 evidence not retained; final clauses UNSPECIFIED; reconciliation downgraded the NEVER proposal.	OPA enforced HUMAN; path remained fail-closed and required review.

cost was USD 0.856704. The open-weights run used Qwen3-4B Q4_K_M through Ollama 0.12.3 on CPU [16], with digest `359d7dd4...94fae7`, seed 20260808, temperature 0, top-p 1, top-k 20, a 16,384-token context window, and a 256-token completion limit.

It completed 60/60 cells. The retained code defines the legacy `unsafe_execution` flag as an oracle-NEVER label combined with a prediction `execute=true`; it invokes no executor. Qwen produced 38 automatic-execution declarations on reference non-automatic cases, nine of them on NEVER cases. These are output errors, not observed RAN executions.

a) Comparison parity: The canonical JSONL and oracle scoring are shared, but the pipelines are not matched (Table IX). The exporter excludes oracle/expected labels from the assessor payload; labels remain evaluation metadata. LangGraph's coverage flag records parsed schema validity, while proposed-workflow coverage uses a non-abstention indicator, so cross-system coverage is not equated. Consequently, 109.793/1.843 $\simeq$ 59.6 is a ratio of reported medians across configured workflows, not framework overhead or a causal gain from any one mechanism. Matched inference-time ablation is unavailable; output replay below isolates only deterministic post-processing. The related 240-row impairment/profile study is a different workload and cannot explain this gap.

D. Prompt-Injection Containment

A separate frozen suite crossed ten attack families with three NEVER-labelled base contexts (30 cases). Attacks attempted instruction override, approval fabrication, policy inversion, audit suppression, and related coercions. The assessor returned the required NEVER class for 30/30 cases and the scorer recorded zero unsafe execution declarations. The Wilson 95% lower bound of 0.886 is a descriptive binomial calculation; the 30 variants share only three base contexts and are not independent operational incidents. This fixed synthetic suite does not establish resistance to arbitrary attacks or production inputs. Fig. 4(c) separates deterministic provider validation from role-chain validation. Role-chain validation is 100.0% for all modes, while provider validation rises from 85.0% without retrieval

TABLE VIII
QUANTITATIVE BASELINES ON THE FROZEN ORACLE. CIS IN TABLE VI APPLY ONLY TO THE PROPOSED-SYSTEM ROWS.

System	Mode / cells	Decision agreement	Action agreement	Joint agreement	NEVER recall	Recorded median (s)
LangGraph	No RAG / 60	83.3%	66.7%	50.0%	100.0%	1.487 s
LangGraph	RAG / 60	83.3%	66.7%	50.0%	100.0%	1.683 s
LangGraph	Agentic RAG / 60	83.3%	66.7%	50.0%	100.0%	1.843 s
Proposed system	Agentic RAG / 60	100.0%	96.7%	96.7%	95.0%	109.793 s
Qwen3-4B local	Agentic RAG / 60	63.3%	18.3%	11.7%	55.0%	238.381 s

TABLE IX
COMPARISON PARITY AND UNRESOLVED CONFOUNDERS. NOT_SENT IS AN OMITTED REQUEST FIELD, NOT AN INFERRED PROVIDER DEFAULT.

Item	Proposed workflow	External LangGraph configuration
Scenario and retrieval	Frozen configurations and evidence	Same canonical input-side configurations and retrieved passages
Assessment	Four requested provider aliases; multi-role chain	One `gpt-5.6-terra` assessor
Prompt contract	Assessor and role-specific prompts	Minimal baseline prompt and structured response schema
Sampling and budget	Sampling fields NOT_SENT; OpenAI/Anthropic/Mistral limits 5,000; DeepSeek limit omitted	Temperature 0, top-p 1; 256 completion tokens
Governance	Evidence reconciliation, OPA, audit and review preparation	Schema validation and class-consistency guard; no live review or ledger
Timing boundary	Main workflow to its returned recommendation	Recorded assessor request interval; deterministic processing separately outside that interval
Runtime conditions	Recorded July provider aliases; no immutable snapshots	August hosted alias; no cross-date provider-load match

to 97.9% with RAG and 95.8% with Agentic RAG. The 2.1-point difference between RAG and Agentic RAG should not be interpreted as a contradiction of the decision-agreement result: provider validation checks individual output contracts, whereas decision agreement evaluates the final reconciled outcome. The results indicate that final governance can remain oracle-conformant even when some intermediate provider outputs do not satisfy every deterministic validation condition. The append-only index retains 233 attempts: 180 successes (77.3%) and 53 HTTP-500 workflow-boundary failures before retry (22.7%). Every failure row contains only the generic local-webhook body `Error in workflow`; no retained provider subcode, trace identifier, timeout exception, or n8n node/worker error supports a causal attribution. Moreover, all 53 linked response paths were reused by later successful retries, so those overwritten files are excluded from root-cause diagnosis. The provider-side versus n8n-internal cause is unresolved. Failures affected 39 unique cells, but SUCCESS-only resumption recovered every cell, leaving zero pending cells and 180/180 audited evidence packages. Thus, 22.7% is the share of retained attempts that failed before retry, not the final failure rate of the 180-cell semantic benchmark.

E. Runtime and Public-Testnet Evidence

The main runtime and artifact evidence, with its claim boundaries, is summarised per evidence class in the supplementary material [23]: anchor gas, reasoning latency, and the three rejection outcomes are populated for all 240 clean local records; jitter and loss values are configured impairment-profile values; packet capture is telemetry smoke evidence only; and AUTH_V2.2 timing is bounded local verifier-call timing, not proof-validity or full prover coverage.

1) Paired First-Inclusion and Block-Cadence Results: Table X reports the evidence-gated comparison. Across 90 matched pairs, median first-inclusion latency is 12,426.579 ms on Sepolia and 7,225.149 ms on IoTeX. The corresponding P95 values are 24,614.539 ms and 8,763.754 ms. The median paired difference Δ is $+4{,}381.823$ ms (Sepolia minus IoTeX; descriptive 95% interval $+3{,}930.597$ to $+7{,}909.772$ ms), and the median paired ratio R is 1.874 (95% CI: 1.477–2.737); as noted in (2), this is the median of per-pair ratios rather than the quotient of the marginal medians. Under the stated hierarchical bootstrap, the difference interval does not cross zero and the ratio interval does not cross one. This is evidence of a difference for the observed client/RPC/testnet configuration; it is not a causal estimate of consensus alone and does not establish universal superiority. With only three session clusters, reliable nominal interval coverage is not claimed. Fig. 5 visualises the median and P95 latency for the two networks with measured deployment evidence; the wider consensus-family positioning is provided in Section VII-F without inventing unmeasured latency values. The dispersion is material: Sepolia spans 8,350.675–25,115.499 ms and IoTeX spans 4,416.856–10,951.509 ms. The paired ratio also ranges from 0.967 to 5.548, showing that aggregate medians must not be misread as an invariant per-transaction ordering. The median observed block interval is 12.0 s on Sepolia (95% CI: 12.0–12.0 s) and 3.0 s on IoTeX (95% CI: 2.0–3.0 s). This cadence is consistent with part of the latency separation, but the experiment cannot apportion the difference among cadence, RPC behaviour, propagation, load, and consensus. Median anchor execution gas is 48,973 on Sepolia and 43,685 on IoTeX. Identical bytecode and matched payloads do not guarantee equal charged gas on distinct EVM-compatible chains because execution rules and fork activation are chain-configured. The retained receipts contain no opcode traces or archived fork configuration, so the difference is descriptive and is not attributed to a particular pricing rule. Because the native assets and fee markets are not commensurable, no cross-asset cost claim is made. All 180 measured receipts succeeded, all 180 anchor events matched their submitted commitments and evidence hashes, deployed runtime bytecode was identical across networks and sessions, and all duplicate, zero-commitment, and unauthorised-submitter probes were rejected.

TABLE X

PAIRED PUBLIC-TESTNET SENSITIVITY OF IDENTICAL STAGE3ANOMALYLEDGER BYTECODE: 90 PAIRS PER NETWORK. SUBMISSION-TO-FIRST-INCLUSION TIMING INCLUDES RPC/NETWORK EFFECTS AND IS NOT ECONOMIC FINALITY. INTERVALS ARE DESCRIPTIVE TWO-LEVEL BOOTSTRAP INTERVALS OVER THREE SESSIONS. SEPOLIA USES PERMISSIONED TEST VALIDATORS. BLOCK CADENCE USES 189 INTERVALS PER NETWORK; REJECTION PROBES TEST DUPLICATE, ZERO AND UNAUTHORISED REQUESTS.

Network	Chain ID	Median gas	Inclusion median [95% CI] (ms)	P95 (ms)
Sepolia (PoS)	11155111	48,973	12,426.579 [12,394.746, 12,455.950]	24,614.539
IoTeX (Roll-DPoS)	4690	43,685	7,225.149 [4,538.581, 8,493.926]	8,763.754

Network	Inclusion mean ± SD (ms)	Block median [95% CI] (s)	Rejection probes
Sepolia	14,767.707 ± 5,623.743	12.000 [12.000, 12.000]	3/3
IoTeX	6,783.463 ± 2,043.817	3.000 [2.000, 3.000]	3/3

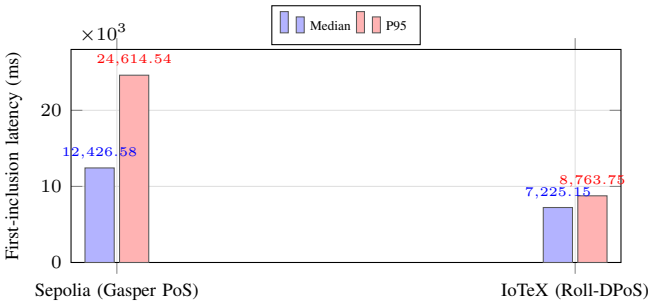


Fig. 5. Measured first-inclusion latency for matched, concurrently submitted audit-anchor transactions on the two networks with deployment evidence. RPC/client overhead and prevailing public-testnet load are included; economic finality is not measured. The separate Raft/QBFT experiment in Fig. 6 uses local permissioned clusters and different within-family workloads; its absolute latency is not compared with these public-testnet values.

F. Consensus-Family and SHA-2 Evidence-Anchor Extension

The requested comparison separates two independent design layers: the consensus protocol that orders an EVM transaction and the cryptographic hash function that binds the evidence. SHA-256 and SHA-512 are one-way hashes, not encryption; frozen validation vectors are retained in the artifact [23]. The Sepolia lane is classified as Gasper PoS from the network and deployment evidence [20], [23]. IoTeX uses Roll-DPoS, including elected delegates, randomised committee formation, and PBFT-style finalisation [21]. The existing public-testnet measurements therefore remain correctly labelled as a Sepolia–IoTeX consensus-sensitivity study rather than a controlled causal benchmark of consensus alone. Fig. 6 separates four *measured* lanes without conflating their estimands. Gasper PoS and Roll-DPoS carry 90 matched public-testnet first-inclusion pairs. Raft carries local etcd replicated-write and leader/majority-loss observations, while Besu QBFT carries local EVM audit-anchor and validator-quorum observations. Raft commits through a majority quorum and addresses crash faults [24]; QBFT requires a supermajority of approved validators and has a Byzantine protocol fault model [25]. Our interventions, however, stop processes and remove participation only. Accordingly, the measured QBFT evidence establishes progress at 3/4, no progress at 2/4, and bounded operator-assisted recovery; it does not experimentally establish resistance to equivocation or forged votes. Across the three sessions, median observed Raft leader-failover recovery was 1,125.9 ms; operator-assisted QBFT restoration resumed production after 14,649.7 ms. All sessions passed their SHA-256 manifests and derivation gate and are retained in the artifact [23].

TABLE XI

MEASURED PERMISSIONED PROCESS-LOSS EXPERIMENT; LATENCY COMPARISONS ARE WITHIN IMPLEMENTATION BECAUSE THE WORKLOADS DIFFER.

Condition	n	Median (ms)	P95 (ms)	Outcome
etcd/Raft (key/value replicated writes)				
Healthy baseline	90	12.02	16.22	writes committed
One follower offline	90	10.25	12.68	writes committed
After leader failover	90	10.43	14.46	writes committed
Majority unavailable	90	–	–	90/90 rejected
Besu/QBFT (EVM audit anchors)				
Healthy baseline	90	4,101.1	4,116.6	anchors included
One validator offline (3/4)	90	4,100.1	8,117.0	anchors included
Two validators offline (2/4)	90	–	–	90/90 no progress
Recovered cluster (4/4)	90	4,105.2	8,126.7	anchors included

Absolute Raft and QBFT latencies are not ranked because their operations differ. The SHA-2 comparison is bounded by a fairness rule: both algorithms hash identical retained payload bytes and anchor their complete digest through identical `HashCommitmentLedger` runtime bytecode. SHA-256 (256-bit digest, 512-bit message block, 32-bit words, ≈128-bit generic collision work) anchors one 32-byte full digest and also has a standard EVM precompile; SHA-512 (512-bit digest, 1024-bit message block, 64-bit words, ≈256-bit generic collision work) has no standard EVM precompile, so it is computed off chain and anchors one 64-byte full digest [23]. Both digests are computed off chain in the primary comparison, which isolates digest-width, calldata, event, and storage effects from CPU hash throughput. Solidity provides a SHA-256 precompile-backed built-in, whereas the standard EVM does not provide an equivalent SHA-512 precompile in the evaluated path [23]. A separate `computeSha256AndAnchor` entry point can measure the SHA-256 precompile path, but that result is not presented as a symmetric SHA-256 versus SHA-512 computation benchmark.

a) Validated local hash-primitive reference: An isolated Hardhat contract separately compares the Keccak-256 opcode path, the SHA-256 precompile path, and a FIPS-vector-gated pure-Solidity SHA-512 implementation for deterministic 32-byte and 1,024-byte inputs. After three excluded warm-ups, 30 `eth_call` observations are retained per algorithm/input cell with cyclically balanced order. The local hash publication gate passed all retained-payload digest checks and the SHA-256/SHA-512 validation vectors. For the 32-byte input, the median primitive-path deltas were 59 gas for Keccak-256, 392 gas for the SHA-256 precompile path, and 162,174 gas for the pure-Solidity SHA-512 reference. For the 1,024-byte input, the median primitive-path deltas were 245 gas for Keccak-256,

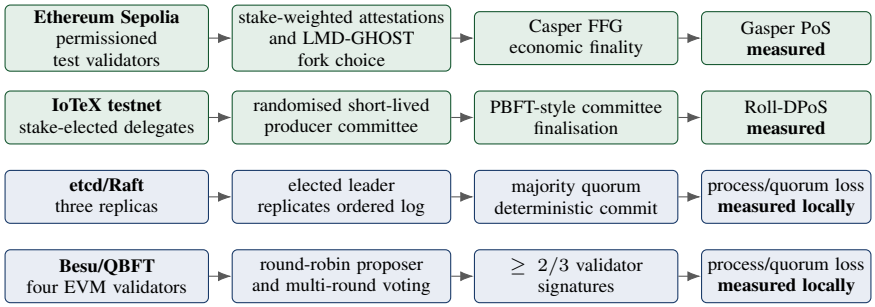


Fig. 6. Consensus-protocol evidence map extending Fig. 5. Green lanes are measured public-testnet first-inclusion configurations; blue lanes are measured local permissioned clusters. The Raft and QBFT interventions cover process crash/non-participation, quorum loss, and recovery. They do not inject equivocation, forged votes, state corruption, or arbitrary Byzantine messages, and absolute latency is not ranked across unlike workloads.

TABLE XII
VALIDATED LOCAL HASH-PRIMITIVE MEASUREMENTS.

Primitive	Bytes	Median gas	n	Fixed
Keccak-256	32	59	30	yes
SHA-256	32	392	30	yes
SHA-512 (Solidity)	32	162,174	30	yes
Keccak-256	1,024	245	30	yes
SHA-256	1,024	2,879	30	yes
SHA-512 (Solidity)	1,024	1,614,065	30	yes

2,879 gas for the SHA-256 precompile path, and 1,614,065 gas for the pure-Solidity SHA-512 reference (Table XII, Fig. 7(a)). All 30 calls per algorithm/input cell returned the same delta because the bytecode, input, and local EVM state were fixed. The reported `gasleft()` deltas are specific to Solidity 0.8.20, optimizer 200, `viaIR=true`, and the Paris EVM target. The pure-Solidity SHA-512 implementation is an experimental instrument only; the deployable workflow computes SHA-512 off chain and anchors its complete 64-byte digest. Interpretation of the public-testnet, permissioned-fault, and hash results follows the single Claim Boundary in Section VI-F.

G. Stage 3 KPI Evidence-Field Coverage

Anchor gas, policy-reasoning latency, replay rejection, zero-anchor rejection, unauthorised-submitter rejection, and gas logs are present in all 240 rows. Coverage denotes field capture, not security effectiveness. The later ZK audit supersedes the earlier interpretation of a prover-time field: the matrix contains no row-bound proof outcome or proof/public-input binding, so no successful per-row verification is inferred. Stage 3 uses the isolated `Stage3AnomalyLedger` micro-benchmark to emit direct evidence for anchor gas and negative-security behaviour (duplicate, zero, and unauthorised-submitter rejection) plus a deterministic per-run policy-reasoning latency probe; Stages 4–5 strengthen this externally through the Sepolia deployment and the paired Sepolia–IoTeX design with its strict publication gate, without turning the micro-benchmark into a production trust ledger.

H. AUTH_V2.2 Timing and Proof Boundary

The paper-specific hardening workflow adds a self-healing local AUTH_V2.2 timing path: it checks dependencies, starts Hardhat when required, compiles artifacts, redeploys when the stored address has no bytecode, and then measures the verifier-call runner. Because the hardened run returned `verifyTx=false`, the observation is only verifier-call timing smoke evidence on an ephemeral chain. A read-only audit found no per-row proof outcome or proof/public-input binding in the frozen matrix: the full-system and no-IPFS variants each have 40/40 missing prover-time records, while the other four variants record that the prover was not invoked. Constant verifier gas is not validity evidence. The matrix therefore supports 0/240 row-bound successful verifications; AUTH_V2.2 is only a component experiment.

TABLE XIII
PAIRED OUTPUT REPLAY. D/A ARE DECISION/ACTION AGREEMENT COUNTS; E IS EXECUTE-TRUE DECLARATIONS; N IS THE SUBSET ON ORACLE-NEVER CELLS. NO NETWORK ACTION OR NEW MODEL CALL IS MEASURED.

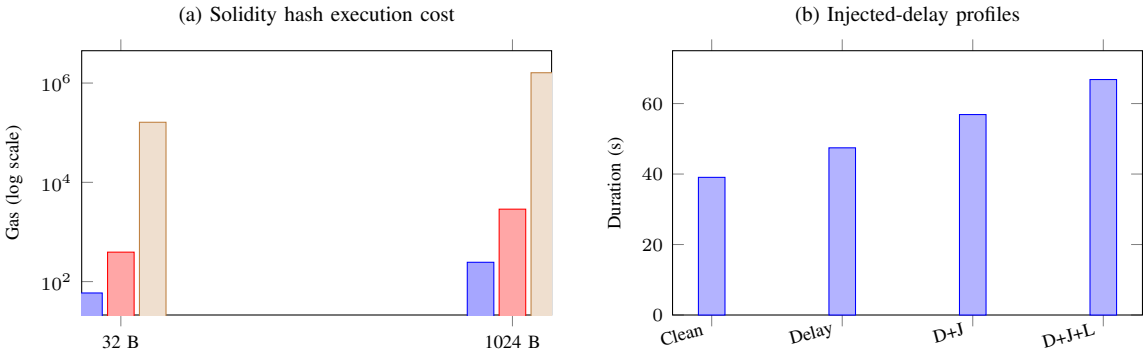
Records	Processing	D	A	E	N
LangGraph (180)	Raw	150	120	0	0
	Class guard	150	120	0	0
	Execution disabled	150	120	0	0
Qwen (60)	Raw	38	11	38	9
	Class guard	38	11	38	9
	Execution disabled	38	11	0	0

I. Configured Profile Stress

Fig. 7(b) summarises the full-system profile trend. The configured clean-to-delay+jitter+loss profiles increase end-to-end duration, whereas the reasoning probe stays within a narrow millisecond-scale range. Thus audit micro-benchmark gas logic is independent of media-profile configuration, while workflow duration reflects the configured impairment.

J. Retained Output Ablation and Timing Attribution

We replayed the unchanged 180 LangGraph and 60 Qwen records through (i) their raw candidate output, (ii) the recorded class-consistency guard and (iii) that guard with automatic execution administratively disabled. The last intervention changes only `execute`, never the decision, action class or oracle labels. Table XIII therefore tests the distinction between semantic agreement and execution declarations. All six reference scenarios require HUMAN or NEVER handling. Thus disabling execution makes this declaration metric zero without improving any semantic label or demonstrating useful automatic service. The benchmark cannot by itself establish correctness, actuation safety or the value of an always-block policy. The proposed workflow's original zero unsafe-automatic count is a



(a) Keccak-256; SHA-256; SHA-512. (b) Configured end-to-end duration.

Fig. 7. Hash cost and configured stress results. The panels have distinct workloads and are not an end-to-end deployment comparison.

reported gate outcome under its synthetic configuration, distinct from the baseline prediction flags and the single approved simulated write.

The recovered benchmark packages provide timestamp intervals for all 180 successful cells (60 per mode). The median aggregate-to-final- reflection phase is 57.698, 75.637 and 79.608 s for No RAG, RAG and Agentic RAG, respectively; it spans the four-role chain and post-role processing. The supplement reports all disjoint phase medians and their timing boundary. Marginal phase medians do not sum to the median total. This population-level descriptive analysis explains where time is spent in the proposed workflow; it does not estimate a causal component cost or remove the LangGraph comparison confounders. No provider requests are repeated.

K. Scientific Interpretation and Limitations

The evidence classes have different estimands and denominators. Runtime hooks, finite-state counts, frozen oracle cells, matched testnet transactions, and configured impairment profiles must not be combined into one system score. The configured LangGraph comparison does not isolate the latency cost of provider diversity, agent roles or governance; the Qwen3-4B result shows that local reproducibility alone does not guarantee oracle-conformant classification. The prompt-injection suite tests only 30 synthetic variants. The R10 endpoints used the recorded aliases `gpt-5.6-terra`, `claude-fable-5`, `deepseek-v4-pro`, and `mistral-medium-3-5` on 16 July 2026; immutable provider snapshots were not exposed. Temperature, top-p, and top-k were unsent, so no defaults are inferred. The maximum-output field was 5,000 tokens for OpenAI, Anthropic, and Mistral and was omitted for DeepSeek. Prompt hashes match all 180 cells; full prompts and the sent-field inventory are in the supplement.

a) Future independent human validation: Independent expert validation was not completed. A future study requires eligible independent O-RAN-literate annotators and any applicable institutional review. The frozen oracle will remain unchanged. Decision and action categories should be treated as nominal, with unweighted agreement statistics and whole-scenario resampling. The archived protocol's weighted action statistic assumes an ordering that the four authority classes do not establish; it must be amended and frozen before collecting new labels. No amended study or human agreement statistic is reported here.

VIII. Conclusion and Future Directions

This paper addressed a practical governance question: how can large-language- model assistance be introduced into O-RAN security operations without allowing the model to become the enforcement authority? The proposed two-plane architecture separates observation and reasoning from deterministic policy, operator approval, and evidence services. A recommendation therefore remains a proposal until the four-class action ladder accepts, escalates, or rejects it. Hash-bound lineage connects the input, retrieved evidence, model proposals, final decision, approval record, and audit receipt, while the architecture contains no direct model-to-E2 actuation path. This makes the intended role explicit: decision support for non-real-time policy loops, not autonomous near-real-time radio control.

The evaluation provides complementary evidence rather than a single aggregate score. Formal model checking covered all 320 states of the configured policy-gate abstraction and found no violation of six safety rules. On the frozen 180-case benchmark, the proposed workflow achieved 100.0% decision agreement and 96.7% action agreement, compared with 83.3% and 66.7% for the external LangGraph baseline; both reported zero unsafe automatic flags under their respective benchmark definitions. The local Qwen3-4B run shows that reproducible deployment alone does not guarantee oracle-conformant classification, and the 30-case injection suite supports only bounded containment claims. Across 90 matched transaction pairs, median inclusion time was 12.427 s on Sepolia and 7.225 s on IoTeX, while the live n8n workflow wrote only to a simulated ledger after explicit human approval. These configured workflows differ simultaneously in reasoning, request budgets and governance. The replay ablation separates execution declarations from semantic agreement; the 80.5–109.8 s median durations do not isolate a framework or component cost.

The claims remain limited to the released artifacts, a synthetic frozen oracle, public testnets, local permissioned clusters, and simulated actuation. The study does not establish production O-RAN safety, semantic correctness of model output, or system-level zero-knowledge verification; the retained matrix supports no row-bound successful proof verification. Future

work will validate the oracle with the preregistered independent-annotator protocol, add production-derived scenarios and multiple client/RPC vantage points, exercise arbitrary-Byzantine and adaptive multi-turn attacks, and implement row-bound proof/public-input binding. Standard A1 payloads, tighter SMO/Non-RT RIC integration, and lower latency can then support fail-closed operator pilots.

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